Extensive-form games

Description

- Extensive form is another way of representing games in a graphical way.
- This form is more suitable for describing games where the moves of the game take place sequentially.
- A game in extensive form is given by a game tree, which consists of a directed graph in which the set of vertices represents positions in the game, and a distinguished vertex, called the root, represents the starting position of the game.

An example

- Recall the battle of the sexes game when the man makes the move, and the woman sees it and then makes her move.
- We may represent it in the following graphical way.

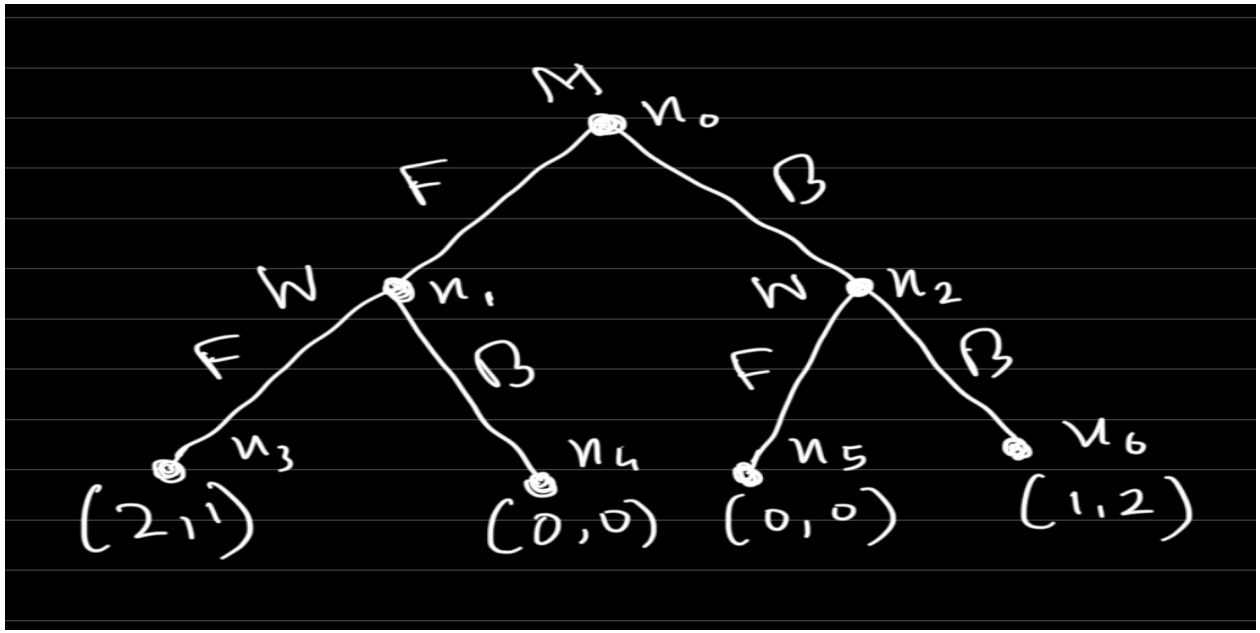


Figure: Figure 1

A directed rooted tree

- A directed rooted tree is a triplet $T = (V, E, x_0)$, where:
  - $\cancel{X}$ V is a finite set with $|V| \geq 2$. The elements of $\cancel{X}$ V are called nodes.
  - E is a subset of $V \times V$. The elements of E are called edges. An edge $e = (x, y) \in E$ is called an outgoing edge of x and an ingoing edge of y.
  - There is an $x_0 \in V$, called the root, such that for each $x \in V \setminus x_0$ there is a unique path from x_0 to x. Here, a path from x_0 to x is a series of edges $(x_0, x_1), (x_1, x_2), \ldots, (x_{k-1}, x_k), (x_k, x)$ for some $k \geq 0$.
  - x_0 has no ingoing edges.

A directed rooted tree

- The nodes that do not have any outgoing edges are called end nodes or terminal nodes or leaves, denoted by $Z(\subseteq V)$.
- For $x \in V$, $C(x)$ denotes the set of nodes, called the children of x, which have incoming edges from x. Formally,

$$C(x) = \{y \mid (x, y) \in E\}.$$

- By $A(x)$, we denote the set of edges which are coming out of x. Formally,

$$A(x) = \{e \in E \mid e = (x, y) \text{ for some } y \in \overset{V}{\cancel{X}}\}.$$

Extensive-form game

- A game in extensive form (or extensive-form game) is a tuple

$$\Gamma = \langle N, V, E, x_0, (V_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

where

$V = \{V_1, V_2, \phi\}$

$V_1 \cup V_2 = V$

  - N is a finite set of players.
  - (V, E, x_0) is a directed rooted tree, called the game tree. $V_1 \cap V_2 = \phi$
  - $(V_i)_{i \in N}$ is a partition of the set of vertices that are not leaves, i.e, $V \setminus Z$.
  - $(u_i)_{i \in N}$ is the collection of utility functions where $u_i : Z \to \mathbb{R}$ for all $i \in N$.

$u = (u_i)_{i \in N}$

$|N| = n$

$u : Z \to \mathbb{R}^n$

Example

- Recall the previous example:

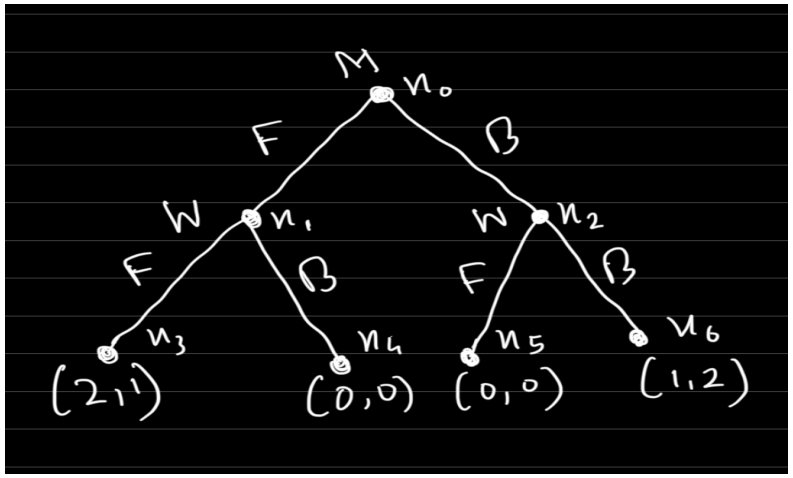


Figure: Figure 2

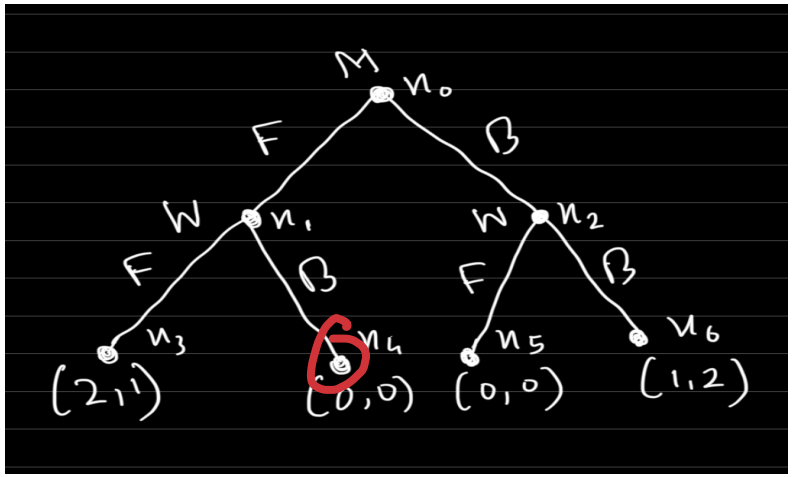
- Here
  - $N = \{M, W\}$
  - $V = \{x_0, x_1, \ldots, x_6\}$.

Example

$v(x_0) = M$

$v(x_1) = W$

$v(x_2) = W$



(x_0, x_1, x_3)

(x_0, x_1, x_4)

(x_0, x_2, x_5)

(x_0, x_2, x_6)

Figure: Figure 2

- $E = \{(x_0, x_1), (x_0, x_2), (x_1, x_3), (x_1, x_4), (x_2, x_5), (x_2, x_6)\}$, and root is x_0.
- $V_M = \{x_0\}$ and $V_W = \{x_1, x_2\}$.
- $u_M(x_3) = 2$, $u_M(x_4) = 0$, $u_M(x_5) = 0$, and $u_M(x_6) = 1$.
- Similarly, $u_W(x_3) = 1$, $u_W(x_4) = 0$, $u_W(x_5) = 0$, $u_W(x_6) = 2$.

How the game proceeds?

- Let $J : V \setminus Z \to N$ such that $J(x) = i$ if $x \in V_i$.
- The game proceeds in the following way:
  - Player $J(x_0)$ initiates the game by choosing a possible action in $A(x_0)$. Equivalently, he chooses an element x_1 in the set $C(x_0)$.
  - If x_1 is not a leaf, Player $J(x_1)$ chooses a possible action in $A(x_1)$ (equivalently, an element $x_2 \in C(x_1)$).
  - The game continues in this manner, until a leaf vertex x is reached, and then the game ends with utilities $u(x)$.

How the game proceeds?

- By definition, the collection of the vertices of the graph is a finite set, so that the game necessarily ends at a leaf, yielding a sequence of vertices $(x_0, x_1, \ldots, x_k)$, where x_0 is the root of the tree, x_k is a leaf, and $x_{l+1} \in C(\overset{x_l}{\cancel{x_0}})$ for $l = 0, 1, \ldots, k-1$.
- This sequence is called a play. Every play ends at a particular leaf x_k with utilities $u(x_k)$.
- Similarly, every leaf x_k determines a unique play, which corresponds to the unique path connecting the root x_0 with x_k.

Perfect information

- In such a game, every player who is to take an action knows the current state of the game, meaning that he knows all the actions in the game that led to the current point in the play.
- This implicit assumption is called perfect information

Strategy

$s_i : V_i \to \bigcup_{x \in V_i} A(x)$

s.t. $s_i(x) \in A(x) \ \forall x \in V_i$

- A strategy is a prescription for how to play a game. The definition is as follows.

Definition

A strategy for player i is a function s_i mapping each vertex $x \in V_i$ to an element in $A(x)$ (equivalently, to an element in $C(x)$).

Example

- Recall the game in the previous example:

$V_M = \{n_0\}$

$s_M : V_M \to A(n_0)$

$s_M(n_0) \in A(n_0) = \{F, B\}$

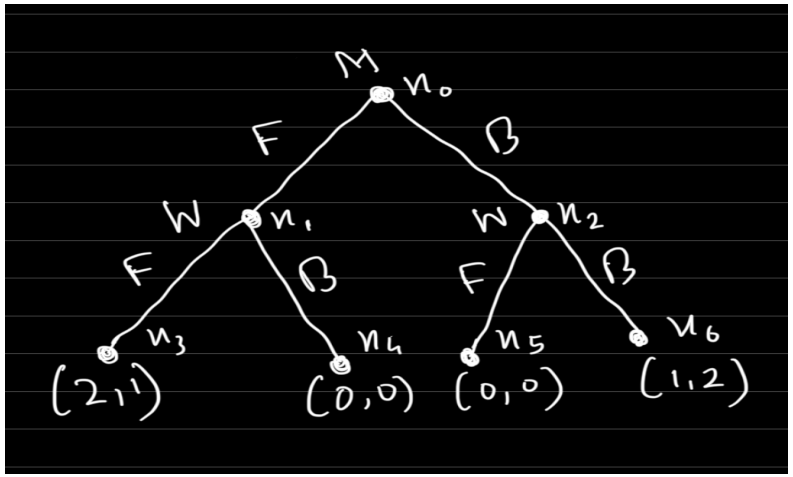


Figure: Figure 2

$V_W = \{n_1, n_2\}$

$s_W : V_W \to \bigcup_{n \in V_W} A(n)$

s.t.

$s_W(n_1) \in A(n_1)$

$s_W(n_2) \in A(n_2)$

- Player M has two strategies $\{F, B\}$, and Player W has four strategies $\{(F, F), (F, B), (B, F), (B, B)\}$.

$(F, (F, F))$

$(F,(F,F)),(B,(B,B))$

Example

- According to this definition, a strategy includes instructions on how to behave at each vertex in the game tree, including vertices that previous actions by the player preclude from being reached.
- For example, consider the following game:

$s_1 : V_1 \to \bigcup_{v \in V_1} A(v)$

if $s_1(v_0) = R$

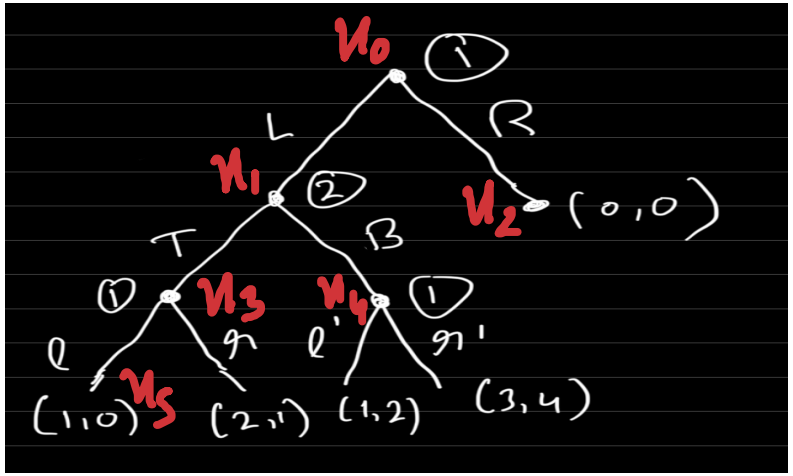


Figure: Figure 3

$s_1(v_0) = R, L$

$s_1(v_3) = l$

$s_1(v_4) = l'$

$s_2(v_1) = T$

(v_0, v_1, v_3, v_5)

$u(s_1, s_2) = (0,0)$

$u(s_1, s_2) = (1,0)$

Example

- For this game, in a strategy of Player 1, even if Player 1 chooses action R at the root node, the strategy must include instructions on how Player 1 should play in the next two nodes even if those two are not reachable because of the choice R.
- The main reason this definition is used is its simplicity: it does not require us to provide details regarding which vertices need to be dealt with in the strategy and which can be ignored.
- We will later see that this definition is also needed for further developments of the theory.

Strategy vector

- A strategy vector is a list of strategies $s = (s_i)_{i \in N}$, one for each player.
- Like before, Player i's set of strategies is denoted by S_i , and the set of all strategy vectors is denoted by $S = S_1 \times S_2 \times \cdots \times S_n$.
- Every strategy vector $s = (s_i)_{i \in N}$ determines a unique play (path from the root to a leaf).
- The play that is determined by a strategy vector $s = (s_i)_{i \in N}$ is $(x_0, x_1, \ldots, x_k)$, where x_1 is the choice of player $J(x_0)$, based on his strategy, x_2 is the choice of player $J(x_1)$ based on his strategy, and so on, and x_k is a leaf.

Strategy vector

$u_1(s_1^*, s_2^*, \ldots, s_n^*)$

$\geq u_1(s_1, s_2^*, \ldots, s_n^*)$

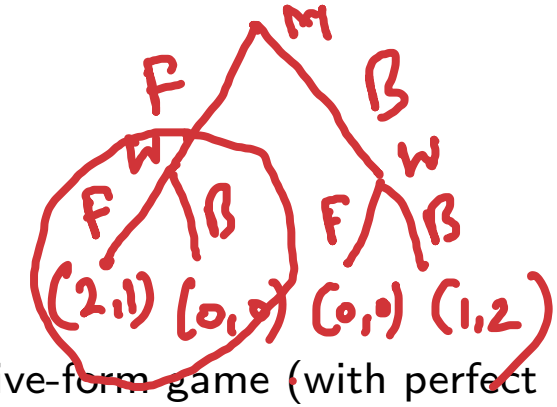
- The play corresponds to the terminal node x_k (with utilities $u(x_k) = (u_1(x_k), \ldots, u_n(x_k))$, which we also denote by $u(s)$.

$s = (s_1, \ldots, s_n)$

$(x_0, \ldots, x_k)$

$u(s) = u(x_k)$

Subgame of a game



Definition

Let $\Gamma = (N, V, E, x_0, (V_i)_{i \in N}, (u_i)_{i \in N})$ be an extensive-form game (with perfect information), and let $x \in V$ be a vertex in the game tree. The subgame starting at x, denoted by $\Gamma(x)$, is the extensive-form game

$\Gamma(x) = (N, V(x), E(x), x, (V_i(x))_{i \in N}, (u_i)_{i \in N})$ where:

- The set of players N is as in the game Γ.
- The set of vertices $V(x)$ includes x, and all the vertices that are descendants of x in the game tree (V, E, x_0), that is, the children of x, their children, the children of these children, and so on.
- The set of edges $E(x)$ includes all the edges in E that connect the vertices in $V(x)$.
- The set of vertices at which Player i is a decision maker is $V_i(x) = V_i \cap V(x)$.
- u_i (with a slight abuse of the notation) is the restriction to the set of leaves in the game tree $(V(x), E(x), x)$.

Subgame of a game

- The original game Γ is itself a subgame of Γ in the sense that $\Gamma(x_0) = \Gamma$.
- Further every leaf x defines a subgame in which no player can make a choice.

Two-player games

- We will discuss games where there are two players and the game may result in either Player 1 winning, or Player 2 winning, or ends in a draw.
- These are two-player zero-sum games with utilities either $(1, -1)$ or $(-1, 1)$ or $(0, 0)$.

Winning strategy and strategy guaranteeing at least a draw

Definition

A strategy s_I of Player I is called a winning strategy if

$$u(s_I, s_{II}) = (1, -1) \text{ for all } s_{II} \in S_{II}.$$

Definition

A strategy s_I of Player I is called a strategy guaranteeing at least a draw if

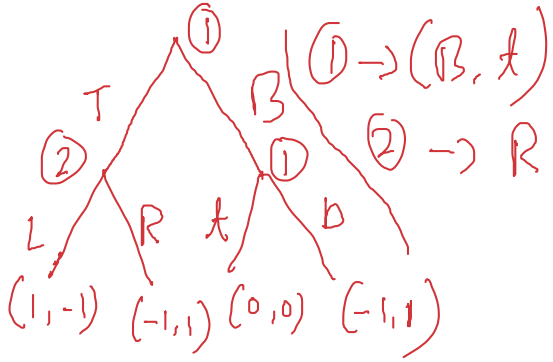
$$u(s_I, s_{II}) \in \{(1, -1), (0, 0)\} \text{ for all } s_{II} \in S_{II}.$$

- A winning strategy for Player II, and a strategy guaranteeing at least a draw for Player II, are defined similarly.

Result

Theorem (Therorem 1, (Zermelo(1913), Von Neumann (1928)))

In every two-player game (with perfect information) in which the set of utilities is $\{(1,-1),(-1,1),(0,0)\}$*, one and only one of the following three alternatives holds:*



1. *Player I has a winning strategy.*
2. *Player II has a winning strategy.*
3. *Each of the two players has a strategy guaranteeing at least a draw.*

Tic-tac-toe

- Tic-tac-toe can be seen as a two-player game (with perfect information) where the set of utilities is $\{(1,-1),(-1,1),(0,0)\}$.
- Therefore, Theorem 1 holds true for Tic-tac-toe.
- Which alternative is true for Tic-tac-toe?

Chomp: David Gale's (1974) game

- It is played on an $n \times m$ board of squares. Each square is denoted by a pair of coordinates (i, j), $1 \leq i \leq n$ and $1 \leq j \leq m$: i is the horizontal coordinate, and j is the vertical coordinate.

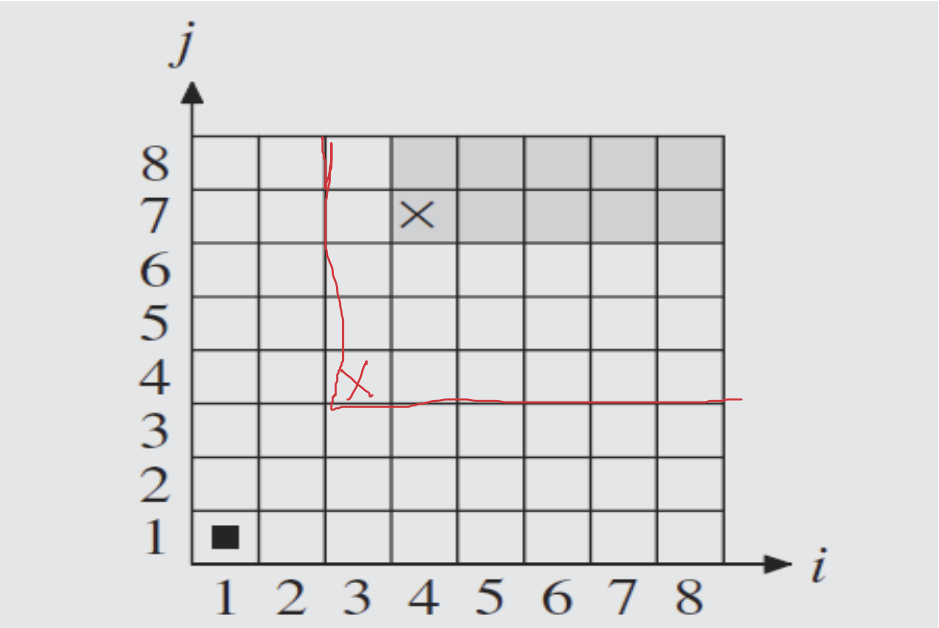
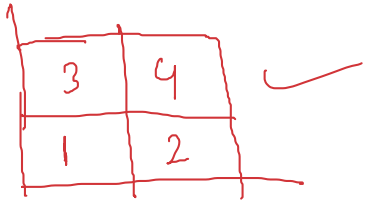
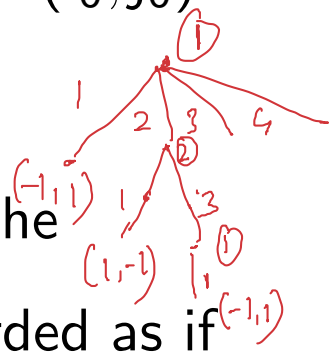


Figure: Figure 4

Chomp: David Gale's (1974) game



- Every player in turn captures a square, subject to the following rules: if at a certain stage the square (i_0, j_0) has been captured, no square that is located north-east of (i_0, j_0) can be captured in subsequent moves.



- This means that after (i_0, j_0) has been captured, all the squares (i, j) satisfying $i \geq i_0$ and $j \geq j_0$ can be regarded as if they have been removed from the board.
- Player I has the opening move. The player who captures square $(1, 1)$ (which is marked in Figure 6 with a black inner square) is declared the loser.

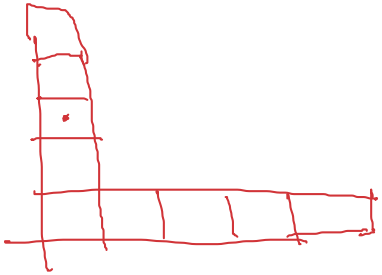
Chomp: David Gale's (1974) game

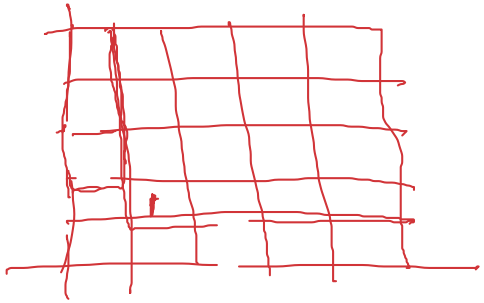
- This game can be represented as a game considered in Theorem 1, and the conclusion of Theorem 1 holds true for this game.
- Which alternative is true for this game?

Chomp: $m = n$

Theorem

In David Gale's game on an $n \times n$ board, the following strategy is a winning strategy for Player I: in the opening move, capture square (2, 2), thus leaving only the squares in row $j = 1$ and column $i = 1$. From that point on, play symmetrically to Player II's actions. That is, if Player II captures square (i, j), Player I captures square (j, i) in the following move.





Chomp: $m = n$

- The above strategy is well defined. That is, if Player II captures square (i, j) (and $(i, j) \neq (1, 1)$), square (j, i) has not yet been removed from the board (verify this!).
- This strategy is also a winning strategy when the board is infinite, $\infty \times \infty$.